ThetaSwap: Adverse Competition Oracle

The Risk Nobody Hedges

In a symmetric equilibrium, each of N LPs earns $1/N$ of fees [Ma and Crapis, 2024]. In practice, sophisticated actors concentrate fee revenue away from passive participants. This is **adverse competition** — a distinct, unhedged risk.

- **Impermanent loss (IL)** — price divergence between pool assets. Variance perpetuals perfectly hedge IL [Fukasawa et al., 2023]; Panoptic deploys this on-chain.
- **Loss-versus-rebalancing (LVR)** — adverse selection cost from informed traders. MEV-aware products address LVR.
- **Fee concentration risk** — the *competition structure* among LPs for fee revenue, captured by the FLAIR metric [Milionis et al., 2024]. **Nothing hedges this today.**

Research Question

Is there empirical demand for a hedging instrument against adverse competition? If so, how much are LPs willing to pay?

Data

ETH/USDC 30bps pool (Uniswap V3 mainnet)
41 days · 600 positions · 3,365 position-day observations · 597 exits

Treatment variable — concentration deviation:

$$\boxed{\Delta^+ = \max\left(0,\ A_T - A_T^{1/N}\right)}$$

- $A_T = \left(\sum_k \theta_k \cdot x_k^2\right)^{1/2}$ fee concentration index (x_k = fee share, $\theta_k = 1/\text{blocklife}$)
- $A_T^{1/N} = \sqrt{\Theta/N^2}$ Ma–Crapis competitive null (equal shares $x_k = 1/N$)
- $\Delta^+ > 0$: fees more concentrated than the symmetric equilibrium

Model: The Shelter Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1 \boxed{\Delta_{t-\ell}} + \beta_2\, \Delta^2_{t-\ell} + \beta_3\, \text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)$$

We expect $\beta_1 < 0$ (shelter effect):

- Mild fee concentration correlates with *high trading volume* — the pool is active, all LPs benefit from elevated fee revenue
- At low Δ, more concentration means more volume, which *lowers* exit probability
- This is **healthy competition**: sophisticated LPs attract flow that benefits everyone [Capponi et al., 2024]
- A linear-only model would conclude concentration is *protective* — missing the adverse regime entirely

Model: The Adverse Competition Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1\,\Delta_{t-\ell} + \beta_2\,\boxed{\Delta^2_{t-\ell}} + \beta_3\,\text{IL}_t + \beta_4\log(\text{age}_{i,t})\big)$$

We expect $\beta_2 > 0$ (adverse competition):

- Beyond a threshold, concentration no longer signals volume — it signals *fee extraction* by sophisticated actors
- The quadratic term Δ^2 captures this **nonlinear regime shift**: the marginal effect of concentration *flips sign* at the turning point
- Together: $\beta_1 < 0 + \beta_2 > 0 =$ **inverted-U** with turning point $\Delta^* = -\beta_1/(2\beta_2)$

Empirical Results

Lag	$\hat{\beta}_1$ (linear)	p	$\hat{\beta}_2$ (quadratic)	p	Turning point
1	$\mathbf{-23.18}$	**0.012**	$\mathbf{+129.20}$	**0.030**	**0.090**
2	-43.42	*0.016*	$+226.92$	*0.065*	*0.096*
3	$\mathbf{-32.44}$	**0.001**	$\mathbf{+205.34}$	**0.004**	**0.079**

For detailed data curation, specification tests, and robustness checks, see `research/model/main.pdf`.

What the Data Says

Passive LPs respond to concentration shocks within one to three days. When fee dispersion exceeds ∼9% above the competitive equilibrium ($\Delta^* = -\beta_1/2\beta_2 = 23.18/(2 \times 129.20) = 0.090$), the shelter effect reverses: each additional percentage point of concentration raises exit probability by approximately 2.3 percentage points ($(\beta_1 + 2\beta_2 \cdot 0.15)\,\bar{P}(1-\bar{P}) = 15.58 \times 0.146 = 2.27$). At a fee dispersion level of 15% above equilibrium, LPs in this pool are willing to pay approximately **10% above the average hourly pool fee revenue** *for protection*
($2.3pp \times 48h$ avg remaining $\approx 1.1h$ of fees, or ∼10% of hourly revenue).

This establishes both the **existence of demand** (the inverted-U with a statistically significant adverse regime) and a **concrete price** (the marginal effect translated to an insurance premium).

Answers

Question	Finding
Is there demand for hedging adverse competition?	**Yes.** LPs react to concentration shocks within days. Below a threshold, concentration is protective; above it, adverse competition drives exits with high statistical significance.
What is the price?	LPs are willing to pay approximately **10% above the average hourly pool fee revenue** for protection — derived from the exit hazard marginal effect at 15% fee dispersion above equilibrium.

Does Hedging Improve Outcomes?

Historical replay of 600 real positions through a power-squared exit payoff ($\alpha = 2$) with \$200K underwriter seed capital:

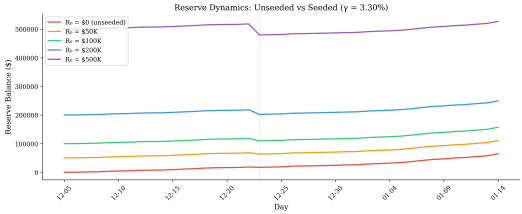
Metric	Result
Positions better off hedged	**18.8%**
Mean hedge value	**+\$23.21** per position
Payout / Premium ratio	121.1% (payouts exceed premiums)
Optimal tail exponent	$\alpha = 2$ (dominates $\alpha = 1, 1.5, 3$)

The power-squared payoff rewards LPs who *stay in the pool* during concentration spikes — no moral hazard.

Backtest: Reserve Dynamics

Reserve trajectories at different seed capital levels show the **bootstrapping problem** and its resolution:

- Without seed capital ($R_0 = 0$): 0% of positions better off — reserve too small at trigger time
- With underwriter seed ($R_0 = \$200K$): reserve absorbs the concentration spike and delivers positive hedge value
- This validates the **hybrid architecture**: the CFMM market layer provides initial liquidity



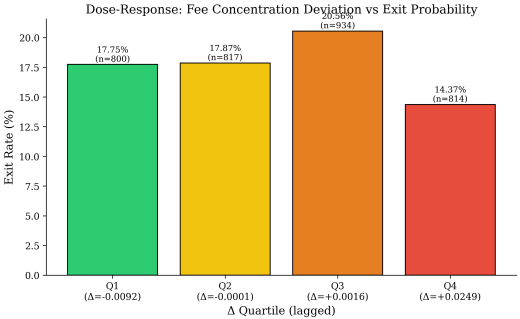
Full backtest methodology and payoff comparison: `notebooks/eth-usdc-backtest.ipynb` and

Tail Risk: The Inverted-U

The dose-response plot reveals the nonlinear structure:

- Exit rates are **stable at baseline** when concentration is at or below the competitive null (Q1–Q2)
- They **spike** at moderate concentration (Q3) — the adverse regime activates
- At extreme concentration (Q4), exits drop — these rare events are driven by volume spikes that benefit all LPs

This is the signature of a **classic insurable tail risk**: rare, severe, and distinct from price-driven losses.

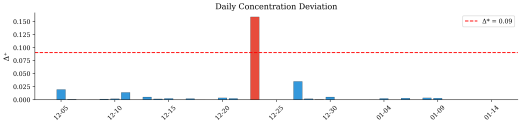


Tail Risk Confirmation

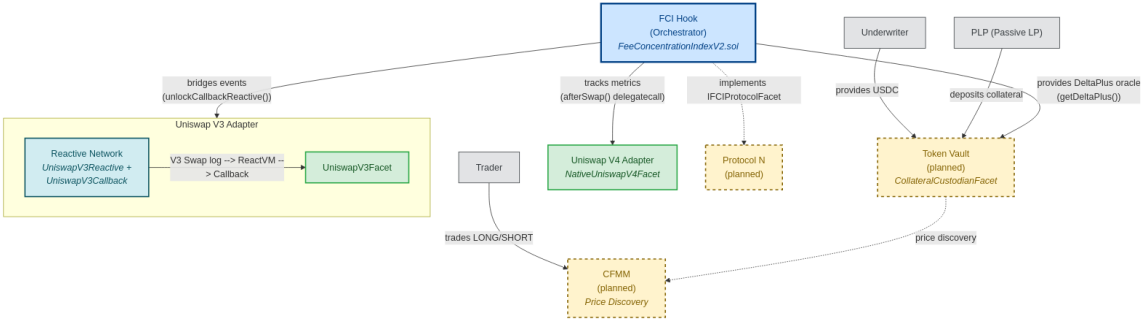
Daily concentration deviation over 41 days:

- 63% of days have $\Delta^+ > 0$ (some concentration)
- **Only 1 day in 41** exceeds the turning point Δ^*
- On that day, Δ^+ jumped from near-zero to **0.158**

Rare but severe — the adverse regime is a tail event. A small, predictable premium protects against low-probability, high-impact concentration spikes.

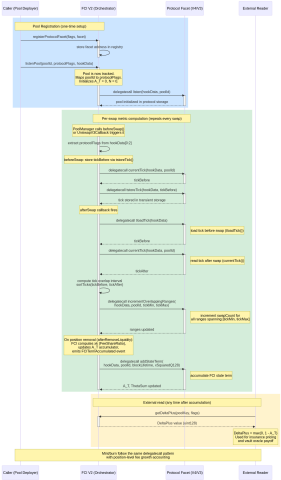


Architecture



FCI Hook at center — protocol adapters (V3, V4) feed swap/mint/burn events, Reactive Network bridges metrics cross-chain, Vault + CFMM settle downstream.

Cross-Protocol Modularity



`listenPool()` → swap/mint/burn events → metric update → Δ^+ derivation.

Live Demo

Run the integration test

```
forge test --match-path
"test/fee-concentration-index-v2/protocols/uniswapV4/
NativeV4FeeConcentrationIndex.integration.t.sol" -vv
```

What to observe:

1. Swap updates A_T — fee concentration index tracks each trade
2. Mint increments N — new positions enter the pool
3. Burn triggers Δ^+ derivation — exit computes concentration deviation
4. All metrics computed on-chain in a single V4 hook callback

What's Next

Built

- FCI oracle live on Uniswap V4
- Reactive adapter bridging V3 events cross-chain
- Backtest validation: 18.8% of positions better off hedged, +$23 mean hedge value

Next

- Linearized power-squared CFMM (trading function complete in spec)
- Vault settlement mechanism

The mathematical specification is complete, the oracle tracks metrics across protocols, and the next milestone delivers the full insurance product.

References I

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M. Fukasawa, B. Maire, and M. Wunsch. Weighted variance swaps and impermanent loss. *arXiv preprint*, 2023. Variance perpetuals perfectly hedge IL; `https://arxiv.org/abs/2212.08323`.

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